

The Frobenius Left-Multiplication Obstruction

Why a Physical Temporal Packet Is Not a Full Ambient Riesz Shell

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Abstract

RH-189–RH-191 formulate an oblique Feshbach problem on the matrix-valued physical orbit. Before computing the complement inverse, one must identify the ambient operator correctly. If $A \in \mathbb{C}^{n \times n}$ and the state is $X \in \mathbb{C}^{n \times m}$, the dynamics is $\mathcal{L}_A X = AX$. Under column-major vectorization,

$$\mathcal{L}_A \simeq I_m \otimes A.$$

Hence

$$\det(zI - \mathcal{L}_A) = \det(zI - A)^m, \quad \text{rank } P_\Gamma(\mathcal{L}_A) = m \text{ rank } P_\Gamma(A).$$

Every base eigenvalue is repeated once per source column. Therefore the literal full-Frobenius Riesz target used informally in the preceding packet discussion is type-wrong: a rank-one root shell or rank-four packet shell cannot equal a complete ambient Riesz projection when $m = 64, 128$, or 256 .

Combined with the exact RH-189 count $N_{\mathcal{L}} = N_K + N_D$, the multiplicity law forces $N_D = m - 1$ around a packet root whenever that root matches one simple base eigenvalue. If it matches no base eigenvalue, the uniform Schur homotopy must fail. A 126-window, 416-root audit finds no complement-free compatible case; the forced one-root complement burden ranges from 63 to 255.

This is a structural correction, not a rejection of the numerical root signal. The source-cyclic space generated by the single matrix S has dimension at most n and need not carry the factor m . It is the correct next setting for a low-rank source-relative spectral channel. Gate A and all Hilbert–Pólya downstream claims remain open.

1 The type hidden by the notation

The RH-185 orbit is written $S, AS, A^2S, \dots$, with $S \in \mathbb{C}^{n \times m}$. Treating each matrix as one Frobenius vector is legitimate, but it does not turn left multiplication into a generic nm -dimensional operator. It preserves the source-column coordinate exactly. This elementary observation changes the Riesz-count target before any numerical inversion is attempted.

Equip

$$\mathcal{H}_F = \mathbb{C}^{n \times m} \tag{1}$$

with the Frobenius inner product $\langle X, Y \rangle_F = \text{tr}(X^*Y)$, and define

$$\mathcal{L}_A X = AX. \quad (2)$$

The issue is not subtle pseudospectral behavior. It is an exact direct-sum decomposition, using standard finite matrix and Riesz-calculus identities [1, 2].

2 Kronecker equivalence

Let $E_j : \mathbb{C}^n \rightarrow \mathcal{H}_F$ place a vector in column j and put zeros in the other columns. Then

$$\mathcal{H}_F = \bigoplus_{j=1}^m E_j \mathbb{C}^n, \quad \mathcal{L}_A E_j = E_j A. \quad (3)$$

The sum is orthogonal in the Frobenius norm.

Theorem 2.1 (Unitary direct-sum form). *Column-major vectorization is unitary from $\mathcal{H}_F$ to $\mathbb{C}^m \otimes \mathbb{C}^n$ and satisfies*

$$\text{vec}(AX) = (I_m \otimes A) \text{vec}(X). \quad (4)$$

Consequently $\mathcal{L}_A$ is unitarily equivalent to the orthogonal direct sum of m copies of A .

Proof. Write $X = [x_1, \dots, x_m]$. Then $AX = [Ax_1, \dots, Ax_m]$, while column-major vectorization stacks precisely these columns. The Frobenius norm is $\|X\|_F^2 = \sum_j \|x_j\|_2^2$, so the identification is unitary. $\square$

Corollary 2.2 (Characteristic and minimal polynomials). *For every $z \in \mathbb{C}$,*

$$\det(zI_{nm} - \mathcal{L}_A) = \det(zI_n - A)^m. \quad (5)$$

The minimal polynomial of $\mathcal{L}_A$ equals the minimal polynomial of A . Thus algebraic multiplicities are multiplied by m , although Jordan-block sizes are not enlarged.

Proof. The determinant of a block diagonal matrix is the product of its block determinants. A polynomial annihilates the direct sum if and only if it annihilates every copy of A . $\square$

The distinction in the last sentence is useful. The obstruction is not an artificial lengthening of Jordan chains; it is an exact replication of every chain across source columns.

3 Resolvents and Riesz projectors

For $z \notin \sigma(A)$, direct calculation gives

$$(zI - \mathcal{L}_A)^{-1}X = (zI - A)^{-1}X. \quad (6)$$

Proposition 3.1 (No resolvent-norm penalty). *In the induced Frobenius norm,*

$$\|(zI - \mathcal{L}_A)^{-1}\|_{F \rightarrow F} = \|(zI - A)^{-1}\|_{2 \rightarrow 2}. \quad (7)$$

Proof. The direct-sum norm is the maximum of the block norms, giving the upper bound. Equality follows by placing a maximizing right singular vector of $(zI - A)^{-1}$ in one column and zeros elsewhere. $\square$

Thus the Frobenius lift does not worsen the resolvent norm. It changes the spectral count.

Let Γ be a positively oriented contour in the resolvent set of A , and let

$$P_\Gamma(A) = \frac{1}{2\pi i} \int_\Gamma (zI - A)^{-1} dz. \quad (8)$$

Theorem 3.2 (Frobenius Riesz multiplicity). *The ambient Riesz projector acts column-wise,*

$$P_\Gamma(\mathcal{L}_A)X = P_\Gamma(A)X, \quad (9)$$

and

$$\boxed{\text{rank } P_\Gamma(\mathcal{L}_A) = m \text{ rank } P_\Gamma(A).} \quad (10)$$

Equivalently, the algebraic eigenvalue count obeys $N_{\mathcal{L}_A}(\Gamma) = mN_A(\Gamma)$.

Proof. Insert (6) into the contour integral. The range is the direct sum of m copies of $\text{Ran } P_\Gamma(A)$, so dimensions multiply. $\square$

4 Consequence for the RH-189 Feshbach count

RH-189 proves that a uniform Schur margin for packet/complement coordinates implies

$$N_{\mathcal{L}_A}(\Gamma) = N_K(\Gamma) + N_D(\Gamma). \quad (11)$$

Combining this with Theorem 3.2 yields an arithmetic law that must be checked before estimating norms.

Corollary 4.1 (Complement congruence). *Under the hypotheses of the RH-189 count theorem,*

$$N_D(\Gamma) = mN_A(\Gamma) - N_K(\Gamma), \quad N_D(\Gamma) \equiv -N_K(\Gamma) \pmod{m}. \quad (12)$$

In particular, a complement-free count is possible only if m divides $N_K(\Gamma)$.

For a contour containing one simple packet eigenvalue, $N_K = 1$. There are only two cases:

1. If $N_A = 0$, equation (12) asks for $N_D = -1$, which is impossible. Hence at least one assumption of the Schur homotopy fails.
2. If $N_A = 1$, the complement count is $N_D = m - 1$. The packet may locate the base eigenvalue, but it cannot be the complete ambient Riesz shell.

The same issue remains for the union of four disjoint root contours. If each contains one simple base eigenvalue, the full ambient count is $4m$, not four, and the complement carries $4(m - 1)$ dimensions.

5 Physical arithmetic audit

The RH-185 data contain 126 temporal windows and 416 packet roots. The matrix-state widths are:

σ	representative m	one-root full count	forced complement
0.04	64	64	63
0.02	128	128	127
0.01	256	256	255

The left and right channels can have different base dimensions n , but they use the same width at a fixed scale. The multiplicity law depends on m , not on n .

No one-root count and no complete length-three or length-four packet count is divisible by its physical width. Thus all 126 literal complement-free targets are arithmetically impossible. This conclusion is exact once the state type is fixed; it does not depend on floating eigenvalue accuracy, contour meshes, or an inverse norm estimate.

6 What survives: the single-seed cyclic space

The direct-sum theorem does not say that every physically relevant subspace has dimension multiplied by m . Let

$$\mathcal{K}_S = \text{span}\{S, AS, A^2S, \dots\} \subset \mathcal{H}_F. \quad (13)$$

This is generated by one vector of the Frobenius Hilbert space. It is $\mathcal{L}_A$ -invariant and has dimension at most the degree of the minimal polynomial of A , hence at most n . Its restriction is cyclic and can have simple eigenvalue counts.

The RH-185 right temporal synthesis lies in $\mathcal{K}_S$ by construction. Thus the numerical packet may still be identifying one source-excited direction from each m -fold ambient eigenspace. The error was to identify that source-relative direction with the complete ambient eigenspace.

An observation seed supplies a further quotient: modes invisible to the input-output moments can be removed. These two reductions are standard in realization theory, but they must be implemented without silently restoring the full-Frobenius count.

7 Revised decision tree

The next calculation should not invert a complement of dimension $nm - L$ and then hope for a rank- L ambient shell. The corrected sequence is:

1. construct the invariant source-cyclic restriction of $\mathcal{L}_A$;
2. prove that the temporal right packet is contained in it and restrict the left functionals to it;
3. identify the source-observable spectral modes selected by the physical observation;

4. only then define packet/complement coordinates and Riesz counts in that reduced state type.

If the temporal roots fail to match the spectrum of the cyclic restriction, the local candidate is rejected. If they match, one obtains a genuine source-channel spectral packet, but not yet a canonical all-level cloud.

8 Boundary of the result

This paper proves a no-go theorem for one literal interpretation only:

$$\text{rank-one/rank-four temporal packet} = \text{complete Riesz shell of } \mathcal{L}_A \text{ on } \mathbb{C}^{n \times m}.$$

It does not reject the RH-185 eigenvalue approximation, a source-cyclic restriction, a controllable/observable minimal realization, or a different physical operator that genuinely mixes source columns.

The distinction is important for the larger program. Gate A requires an intrinsic and transportable physical spectral interface. Correcting the ambient multiplicity is necessary for that interface, but it does not construct it. Gates B–E, a self-adjoint Hilbert–Pólya operator, zeta-zero identification, and the Riemann hypothesis are untouched.

9 Reproducibility

The accompanying implementation tests the Kronecker action, determinant power, resolvent norm, and count arithmetic on independent matrices. The physical audit reads the frozen RH-185 records and stores every width, ambient dimension, packet length, and forced complement burden. It contains no fitted threshold and no spectral matching step.

The full command sequence is recorded in the README. A publication manifest hashes source, tests, frozen inputs referenced by name, result JSON, and the compiled paper.

References

- [1] Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.
- [2] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, second edition, 1995.